Health360 Tableau Project Documentation

Project overview

This project focuses on developing an interactive Tableau-based healthcare analytics dashboard named Health360. The objective is to transform raw healthcare data into meaningful visual insight that supports clinical review, lifestyle analysis, and public health planning. The dashboard is designed to help users identify patterns in chronic disease risk, evaluate lifestyle behaviors, and assess community health trends without relying on manual spreadsheet analysis.

1. Project objective

The main goal of the project is to create an end-to-end analytics solution that converts a fictional health dataset into a dashboard with clear business value. The solution is intended to support three scenarios: clinical risk assessment, healthcare data analysis, and public health research. Each scenario is addressed through dashboards that focus on key indicators such as BMI, blood pressure, cholesterol, glucose, exercise level, smoking status, alcohol use, and age-based disease patterns.

2. Requirements and scenarios

The project was built around three required user personas and corresponding business questions. Scenario 1 is for Dr. Michael Carter, who needs early identification of high-risk patients based on BMI, blood pressure, cholesterol, and glucose patterns. Scenario 2 is for Emily Rodriguez, who explores how lifestyle factors such as smoking, alcohol use, and exercise levels relate to chronic disease conditions. Scenario 3 is for Dr. Priya Nair, who studies community health trends and aims to find patterns in disease counts, BMI categories, symptom reporting, and exercise behavior.

3. Data understanding

The dataset used in the project is a fictional sample dataset containing participant-level records. It includes fields such as participant ID, age, BMI category, exercise level, chronic condition, primary symptom, systolic blood pressure, cholesterol, glucose, heart disease, diabetes, stroke, smoking status, and alcohol use. The dataset is intended for demonstration and educational use only. It is not a clinical dataset and should not be used for diagnosis or treatment decisions.

4. Data preparation and cleaning

The raw CSV file was reviewed to confirm field names, categories, and data consistency. Categorical values such as BMI category, exercise level, smoking status, and alcohol use were standardized for Tableau analysis. Numerical columns such as age, blood pressure, cholesterol, and glucose were validated to ensure proper data types were used. The dataset was prepared using a clean and consistent structure suitable for visual analysis.

5. Tableau workflow and implementation

The project was implemented in Tableau using a structured workflow. The process began with loading the prepared CSV file into Tableau and validating the data fields. After understanding the dataset, a Hyper extract was created to improve performance and enable efficient dashboard interactions. Visualization components were then added to represent clinical measures, lifestyle patterns, and community health trends. Finally, dashboard formatting and filters were applied to make the workbook easy to interpret and present.

6. Dashboard design and key visualizations

The dashboard includes several core visualizations that align with the project scenarios. These charts highlight average BMI, average health risk factors, exercise levels, chronic disease count by age group, BMI distribution, diabetes prevalence by age, and smoking status patterns. Each chart is designed to answer a specific analytical question and support quick decision-making for users such as clinicians, analysts, and researchers.

7. Scenario-based insights

Scenario 1 uses the dashboard to support early identification of high-risk patients by comparing BMI, blood pressure, cholesterol, and glucose across groups. Scenario 2 helps analysts connect lifestyle habits with disease patterns and health outcomes. Scenario 3 provides a community health lens by comparing age groups, BMI categories, exercise habits, and recurring symptoms. The dashboard does not imply causation, but it highlights patterns that can help guide deeper research, preventive action, and targeted health interventions.

8. Project outcome

The final output of the project includes the Tableau workbook file and a dashboard image for presentation purposes. These files demonstrate the completion of the required project deliverables and the successful transformation of raw healthcare data into a decision-support tool. The workbook is structured to support clear storytelling, visual exploration, and evidence-based interpretation of public health patterns.

9. Conclusion

The Health360 Tableau project successfully demonstrates the use of business intelligence in healthcare analytics. By combining data analytics with interactive dashboards, the project helps stakeholders understand disease risk, detect trends, and support preventive healthcare planning.